

Total No. of Questions : 8]

SEAT No. :

PD4557

[Total No. of Pages : 2

[6404]-61

B.E. (Civil)

AIR POLLUTION AND CONTROL

(2019 Pattern) (Semester - VII) (Elective - IV) (401004A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume any other data, if necessary.

- Q1)** a) Enlist Various methods of Sampling Particulate matter and Explain any one in detail. [7]
- b) Describe WHO, Air Quality guidelines 2021. [5]
- c) Describe the Important functions of CPCB. [5]

OR

- Q2)** a) Explain with neat sketch location of sampling ports and traverse point. [7]
- b) Write a short note on high volume sampler. [5]
- c) What is iso kinetic, sub iso kinetic iso kinetic sampling? Why is it required? [5]

- Q3)** a) Describe the importance of Air Quality Modelling? [7]
- b) Explain the strength and limitation of clear ACT(1970) [5]
- c) Describe the responsibilities of Environmental Protection agencies. [5]

OR

- Q4)** a) State Classification of Air quality Modelling. [7]
- b) Write short note on USEPA. [5]
- c) Write short note on EPA. [5]

P.T.O.

- Q5) a)** A cylindrical Electrostatic precipitator of Diameter 0.3 m is used for Separating pulverized coal fly ash particles from a furnace gas stream. If volumetric flow rate of the gas is $0.05 \text{ m}^3/\text{s}$, what will be the length of the precipitator for obtaining a collection efficiency of 99.9%? what percent change in electrode collection area is required to increase the collection efficiency from 99.9 to 99.95%. [8]
- b) Write note on Fabric Filters. [5]
- c) What are the incinerators? Draw sketch and explain its working. [5]

OR

- Q6) a)** A monodispersed aerosol $1.99 \mu\text{m}$ in diameter is passed through a gravity settler 20 cm wide and 50 cm long with 18 plates and channel thickness 0.124 cm. the gas flow is rate 8.6 LPM and it is observed that it operates at an efficiency of 64.9% how many plates would be required to have the unit operate at 80% efficiency. [8]
- b) Write a note on flood and ducts. [5]
- c) Describe advantage and disadvantages of gravity settling chamber. [5]
- Q7) a)** Define indoor air pollution, state causes and effects of indoor air pollutants. [6]
- b) Explain Sick building Syndrome. [6]
- c) Explain air cleaning system for indoor air pollution control. [6]

OR

- Q8) a)** How can the indoor air pollution be controlled using ventilation? [6]
- b) Explain method of Measurements of odor. [6]
- c) Describe the method of oxidation used to eliminate odor. [6]

